MDIS Tashkent

GROUP ORAL PRESENTATION

MALARIA

AN ANCIENT PARASITE, A MODERN FIGHT



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PLAN OF THE TALK

OUTLINE

01

WHAT IS MALARIA

Definition, cause and fatality rate

◆ ◆ ◆

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02

HISTORY

First records, discoveries, major outbreaks

◆ ◆ ◆

Muhammadrizo Asatullayev

03

MALARIA TODAY

Statistics on cases, deaths and damage

◆ ◆ ◆

Ramazan Bakiyev

04

FIGHTING BACK

Vaccine, prevention and treatment

◆ ◆ ◆

Akmalbek Abduraimov

05

CONCLUSION

Key takeaways

◆ ◆ ◆

Javohir Azamov

WHAT IS MALARIA?

> *“A life-threatening disease spread to humans by some types of mosquitoes. The infection is caused by a parasite and does not spread from person to person.”*
>
> WORLD HEALTH ORGANIZATION



CAUSE

A parasite

Plasmodium, a single-celled organism. Not a virus, not a bacterium.

CARRIER

Female Anopheles

The mosquito bites mostly at night. Parasites go to the liver, then to red blood cells.

SPECIES

5 infect humans

P. falciparum is the deadliest. P. vivax is the most common outside Africa.

Source: World Health Organization (2025a)

HOW DEADLY IS IT?

10–15
DAYS

until the first symptoms: fever, chills and headache. It looks like the flu.

24
HOURS

Left untreated, P. falciparum malaria can progress to severe illness and death.

≈100%
UNTREATED

Severe malaria without treatment: deaths approach 100%. With treatment: 10–20%.

*The key word is **TIME**: fast diagnosis saves lives.*



Source: World Health Organization (2025a); World Health Organization (2015)

FIRST RECORDS

CHINA

c. 2700 BC

Fevers that keep coming back

A Chinese medical text describes repeating fevers and an enlarged spleen.

EGYPT

1570 BC

Papyri and a pharaoh

Egyptian papyri describe fevers. In 2010, parasite DNA was found in Tutankhamun’s mummy.

GREECE

c. 400 BC

Hippocrates and the marshes

He links the fevers to life near marshes and stagnant water.

ITALY

THE NAME

Mal aria = “bad air”

People blamed the air of the swamps, and the disease got its name.

Source: Cox (2010); Hawass et al. (2010)

THE GREAT DISCOVERIES



1880

ALPHONSE LAVERAN

A French army doctor in Algeria sees the parasite in the blood of a sick soldier.

NOBEL PRIZE 1907



1897

RONALD ROSS

In India, he finds the parasite inside a mosquito: 20 August, “Mosquito Day”.

NOBEL PRIZE 1902



1898

GIOVANNI BATTISTA GRASSI

His Italian team proves that Anopheles mosquitoes carry human malaria.

ITALY · 1898

Source: Cox (2010); Nobel Prize Outreach (2026). Portraits: Wellcome Collection, public domain

MAJOR OUTBREAKS



80,000

DEATHS

CEYLON (SRI LANKA) · 1934–35

After a drought, about 1.5 million people fell ill.

150,000

DEATHS

ETHIOPIA · 1958

About 3 million cases in one epidemic.

×4

CASES

PAKISTAN · 2022

After the floods: 400,000 cases (2021) → 1.6 million+ (2022).

+5 M

NEW CASES

3 COUNTRIES · 2024

Ethiopia, Madagascar and Yemen added them in one year.

Source: Gill (1935); Fontaine, Najjar and Prince (1961); World Health Organization (2023a; 2025b)

MALARIA TODAY

282 MILLION

malaria cases in 2024, in 80 countries



610,000

DEATHS IN 2024

almost 1,700 people every day

94% · 95%

AFRICA

of all cases · of all deaths

3 OF 4

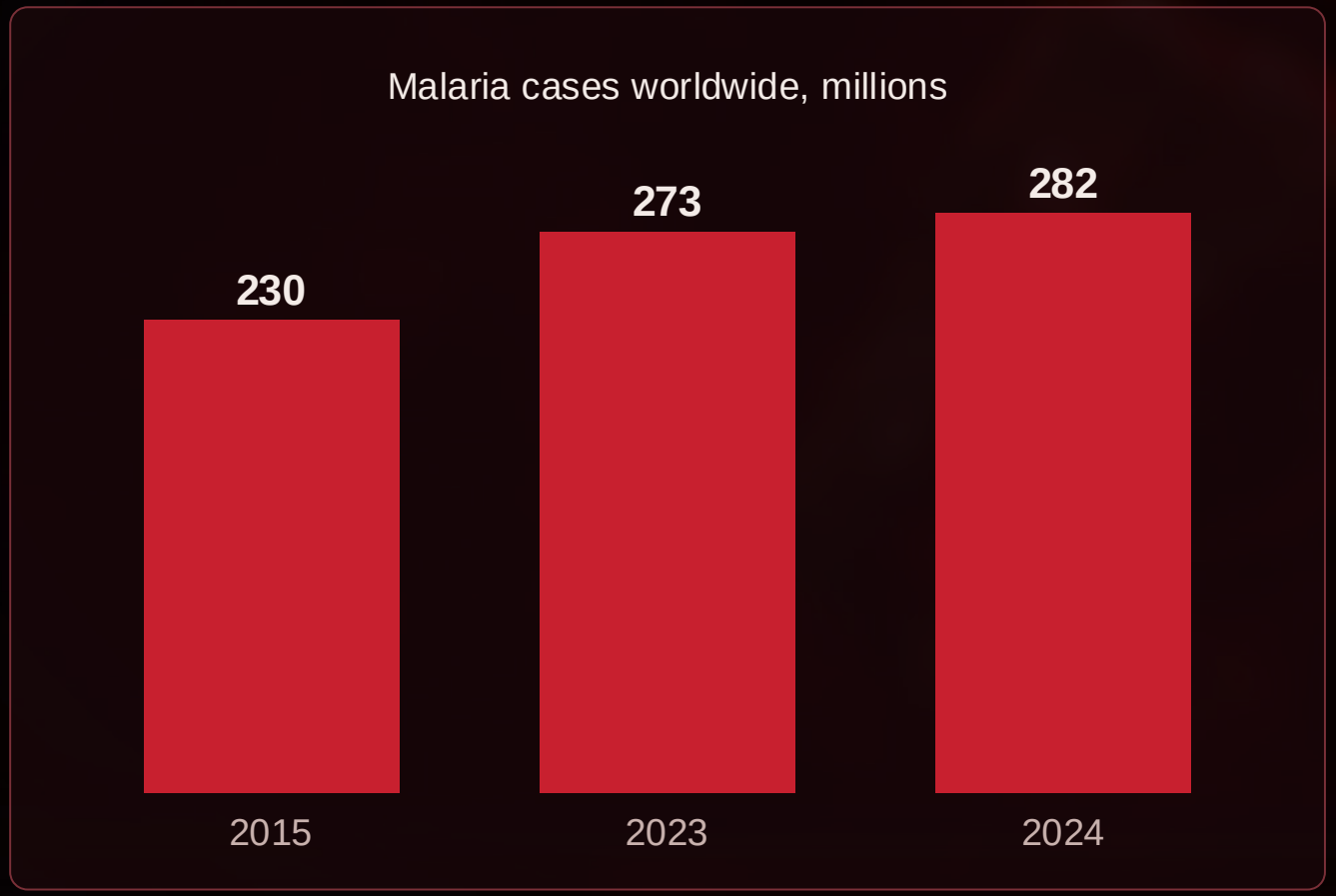
CHILDREN UNDER 5

of all malaria deaths in Africa

5 countries have more than half of all cases: **Nigeria · DR Congo · Uganda · Ethiopia · Mozambique**

Source: World Health Organization (2025b), World malaria report 2025

THE NUMBERS ARE RISING

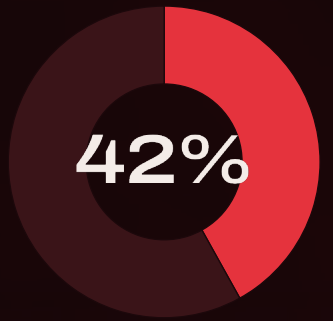


DEATHS · 2015 → 2024

578,000 → 610,000

The death rate per person fell, but the population keeps growing.

FUNDING IN 2024



US$ 3.9 bn
invested
US$ 9.3 bn
needed every year

Source: World Health Organization (2025b)

MALARIA CAN BE DEFEATED

2.3 BN

CASES PREVENTED

since 2000

14 M

DEATHS PREVENTED

since 2000

47 + 1

MALARIA-FREE

countries + territory, WHO-certified



UZBEKISTAN: OUR OWN SUCCESS STORY

1961	2005	2010	2018
First elimination	Tashkent Declaration	Last local cases	WHO certificate, 7 Dec

Surkhandarya, the most affected province: **81 cases in 2007 → 0**

Source: World Health Organization (2018); World Health Organization (2025b)

THE VACCINE

FIRST · WHO 2021

RTS,S

AS01 · GSK

30 years of research. Recommended for children in October 2021: “a historic moment”.

SECOND · WHO 2023

R21

MATRIX-M · OXFORD

−75% cases where malaria is seasonal. Cost-effective at US$ 2–4 per dose.

−13% child deaths from all causes in the pilot: Ghana, Kenya, Malawi

24 countries using malaria vaccines by October 2025



Source: World Health Organization (2021; 2023b; 2025b); University of Oxford (2023)

PREVENTION

1 · INSECTICIDE-TREATED BED NETS

68% of all cases prevented in Africa, 2000–2015. Sleep under it every night.

2 · INDOOR SPRAYING

Insecticide on the inside walls, where mosquitoes rest after they bite.

3 · SEASONAL MEDICINES FOR CHILDREN

About 54 million African children in each round, 2024.

4 · PERSONAL PROTECTION

Repellent, long sleeves, no standing water. Preventive tablets for travellers.



Source: Bhatt et al. (2015); World Health Organization (2025a; 2025b)

TREATMENT

STEP 1 · TEST FIRST microscope or rapid diagnostic test

1600s Cinchona bark from South America is used against fever.

1820 Scientists extract quinine from the bark.

1970s Tu Youyou finds artemisinin in sweet wormwood. Nobel Prize 2015.

TODAY ACT: artemisinin-based combination therapy. Severe malaria: artesunate injection.



NEW THREAT: DRUG RESISTANCE

Partial resistance to artemisinin is confirmed in Eritrea, Rwanda, Uganda and Tanzania.

Source: Achan et al. (2011); World Health Organization (2015; 2025b); Nobel Prize Outreach (2026)

CONCLUSION

01

A parasite, not a virus or a bacterium. Without treatment it can kill within 24 hours.

02

Almost 5,000 years with humanity, from ancient China to Tutankhamun.

03

282 million cases, 610,000 deaths in 2024, mostly young children in Africa.

04

We know how to win: vaccines, bed nets and fast treatment. Uzbekistan is the proof.

BLOCK THE BITE · TEST EARLY · TREAT FAST

Malaria is preventable and curable.

MDIS Tashkent

THANK YOU

FOR YOUR ATTENTION

QUESTIONS ARE WELCOME



Javohir Azamov · Muhammadrizo Asatullayev · Ramazan Bakiyev · Akmalbek Abduraimov

SOURCES · HARVARD STYLE

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